Networks and Telecommunications

TYPES OF COMPUTER NETWORKS, WIRED & WIRELESS NETWORKS, INTRANET AND THE INTERNET

Networks and Telecommunications

- It is a field that deals with the design, implementation, and management of computer networks and telecommunications systems
- It encompasses various aspects, including
  - Network architecture and protocols
  - Network security and cryptography
  - Telecommuncation systems and technologies (wired and wireless)
  - Data transmission and switching
  - Network performance analysis and optimization
  - Network management and troubleshooting

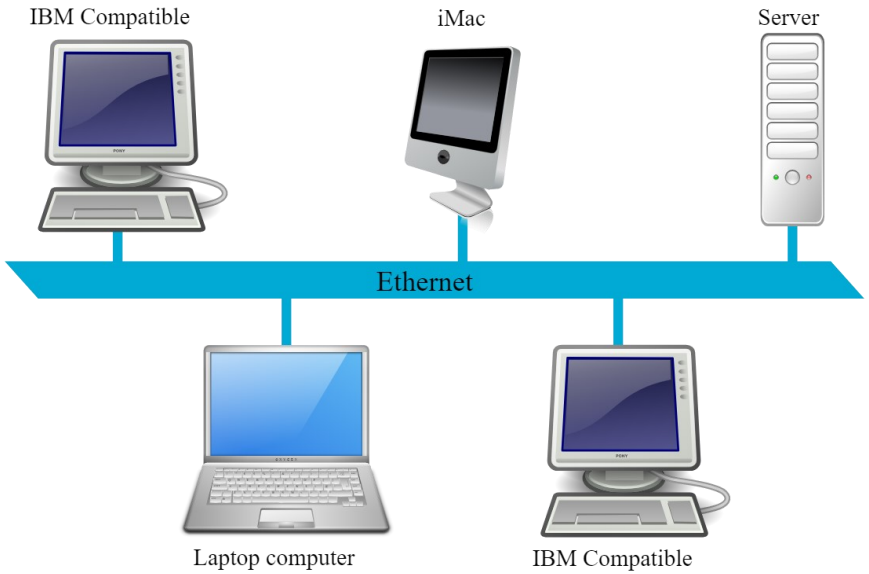
Networks and Telecommunications

- Telecommunications systems enable the transmission of data over public or private networks.
- A network is a communications, data exchange and resource sharing system created by linking two or more computers and establishing standards, or protocols, so that they can work together.

Types of computer networks

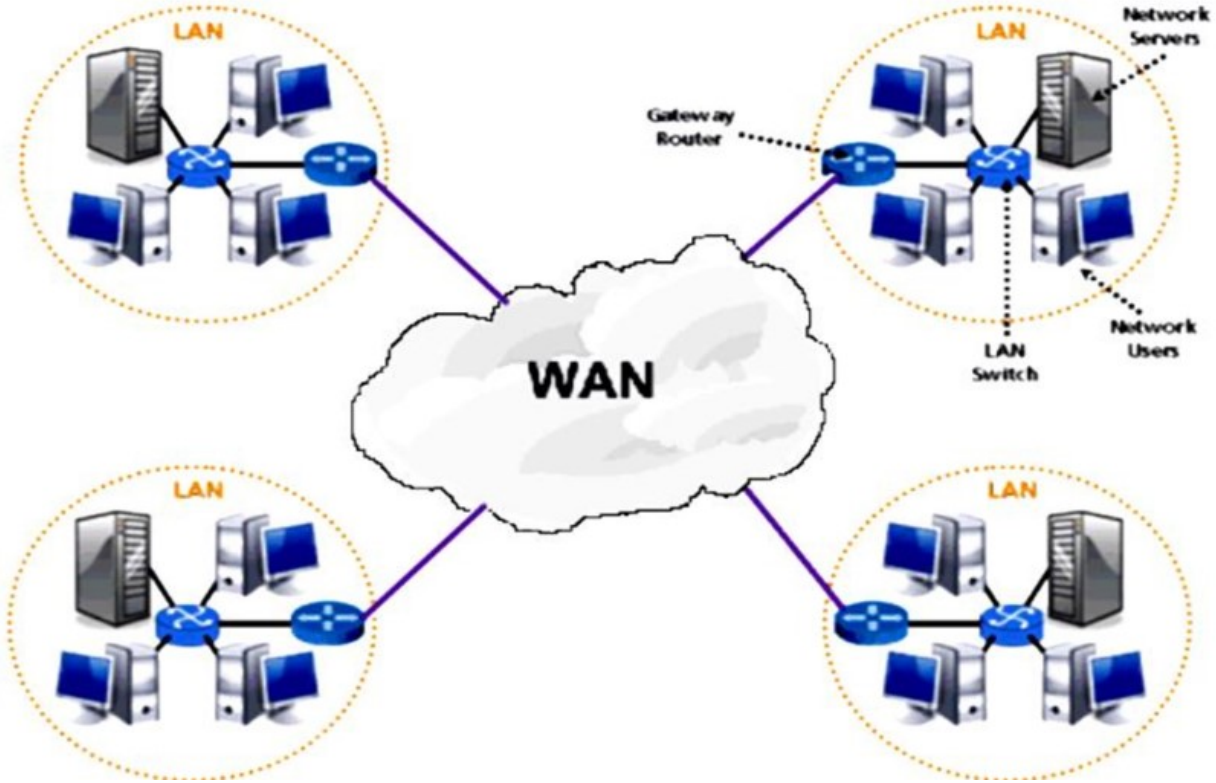
Local Area Network (LAN)

- It is a network that uses cables or radio signals to link two or more computers within a geographically limited area, generally one building or a group of buildings. A network office building, school, or home usually contains a single LAN. The linked computers are called workstations.



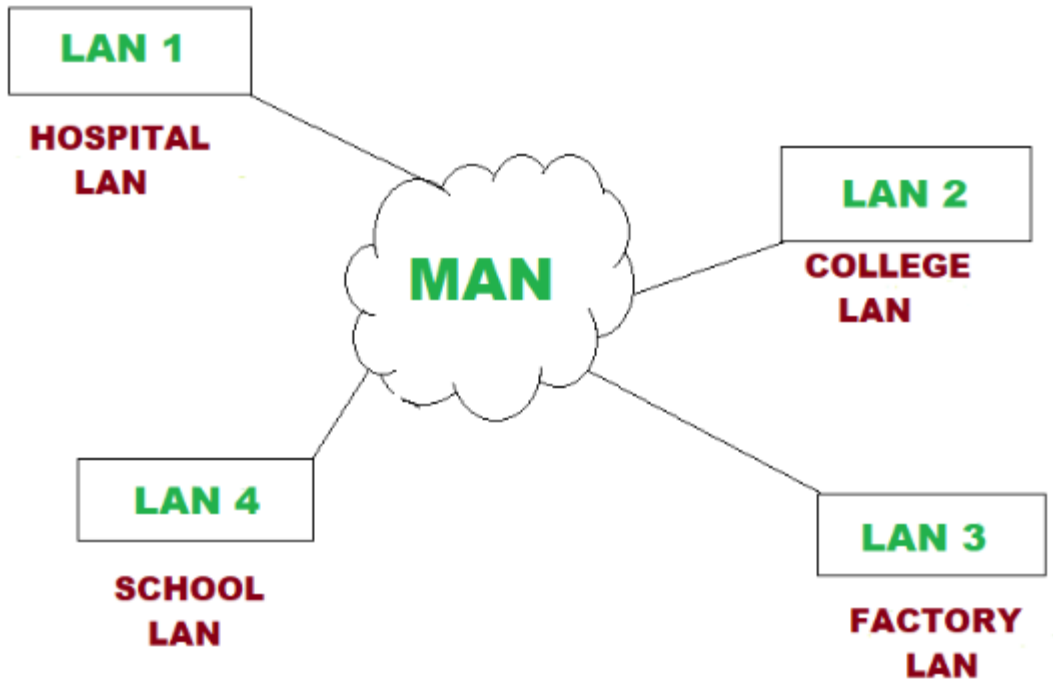
Types of computer networks

- **Wide Area Network (WAN):** a network that provides data communication services for business in geographically dispersed areas such as across a country or around the world). The Internet is a WAN that spans the world.



Types of computer networks

- **Metropolitan Area Network (MAN):** a network that provides a connectivity in a geographic area or region larger than that covered by a local area network, but smaller than the area covered a wide area network. A college or business may have a MAN that joins the different LANs across its campus.



Wireless and Wired Networks

Wireless networks or Wireless Fedelity (wi-fi)

- It is a means of linking computers using infrared or radio signals.
- It is a type of Ethernet, which makes the wireless network a straightforward extension of the wired network.
- These signals are limited in range - usually up to 50 metres - but are subject to interference.
- They can also be blocked by walls. However, they are ideal for mobile devices as a device can connect to a network as long as it is within the range of a wireless access point (WAP).

Wired network

- A wired network connects devices using physical cables, such as Ethernet cables, to transmit data.
- This type of network is often used in:
  1. Home networks: Connecting devices like computers, smart TVs, and gaming consoles to a router.



Wired network contd/

2. Business networks: Linking computers, servers, and other devices in an office setting.
3. Internet service providers (ISPs): Delivering internet connectivity to customers through cable or fiber-optic connections.

Wired network contd/

- Wired networks offer:
  - ✓ Faster data transfer speeds
  - ✓ Greater reliability and stability
  - ✓ Improved security (less vulnerable to hacking).
- However, they can be:
  - ✓ Less flexible (cables can be cumbersome)
  - ✓ More difficult to set up and modify
- Common wired network technologies include Ethernet (RJ-45) and Fiber Optic connections.

Intranet and Internet

- The internet and intranet are two types of networks that serve different purposes.
- **Internet**
  - ✓ Is a global public network connecting millions of devices worldwide.
  - ✓ It allows communication and data exchange between devices across the globe
  - ✓ Accessible to anyone with an internet connection.
  - ✓ Uses standardized protocols like TCP/IP to facilitate communication.
  - ✓ Examples of the internet, World Wide Web (WWW), online services, social media, email etc.

Intranet and Internet contd/

- The internet and intranet are two types of networks that serve different purposes.
- **Intranet**
  - ✓ Is a private, internal network within an organisation or entity.
  - ✓ It has restricted access to authorised personnel or devices.
  - ✓ Used for sharing information, resources, and services within the organisation.
  - ✓ May use similar protocols as the internet but is isolated from public access.
  - ✓ Examples: Company networks, university networks, government networks.

Key difference between Intranet and Internet

- Access: Internet is public, while intranet is private.
- Purpose: Internet connects the world, while intranet serves internal communication.
- Security: Intranet is generally more secure due to restricted access.

Internet Vocabulary

- Networking communication is full of some very technical concepts based on some simple principles. Learn the terms below and you'll be able to hold your own in a conversation about the Internet.
  - ✓ Packet: The fundamental unit of data transmitted over the Internet. When a device intends to send a message to another device (for example, your PC sends a request to YouTube to open a video), it breaks the message down into smaller pieces, called packets. Each packet has the sender's address, the destination address, a sequence number, and a piece of the overall message to be sent.
  - ✓ Hub: A simple network device that connects other devices to the network and sends packets to all the devices connected to it.

Internet Vocabulary contd/

- ✓ Hub: A simple network device that connects other devices to the network and sends packets to all the devices connected to it.
- ✓ Bridge: A network device that connects two networks together and only allows packets through that are needed.
- ✓ Switch: A network device that connects multiple devices together and filters packets based on their destination within the connected devices.
- ✓ Router: A device that receives and analyzes packets and then routes them towards their destination. In some cases, a router will send a packet to another router; in other cases, it will send it directly to its destination.

Internet Vocabulary contd/

- ✓ IP Address: Every device that communicates on the Internet, whether it be a personal computer, a tablet, a smartphone, or anything else, is assigned a unique identifying number called an IP (Internet Protocol) address. Historically, the IP-address standard used has been IPv4 (version 4), which has the format of four numbers between 0 and 255 separated by a period. For example, the domain Saylor.org has the IP address of 107.23.196.166. The IPv4 standard has a limit of 4,294,967,296 possible addresses. As the use of the Internet has proliferated, the number of IP addresses needed has grown to the point where the use of IPv4 addresses will be exhausted. This has led to the new IPv6 standard, which is currently being phased in. The IPv6 standard is formatted as eight groups of four hexadecimal digits, such as 2001:0db8:85a3:0042:1000:8a2e:0370:7334. The IPv6 standard has a limit of 3.4×1038 possible addresses. For more detail about the new IPv6 standard, see this Wikipedia article.

Internet Vocabulary contd/

- ✓ Domain name: If you had to try to remember the IP address of every web server you wanted to access, the Internet would not be nearly as easy to use. A domain name is a human-friendly name for a device on the Internet. These names generally consist of a descriptive text followed by the top-level domain (TLD). For example, Wikepedia's domain name is wikipedia.org; wikipedia describes the organization and .org is the top-level domain. In this case, the .org TLD is designed for nonprofit organizations. Other well-known TLDs include .com, .net, and .gov. For a complete list and description of domain names, see this Wikipedia article.
- ✓ DNS: DNS stands for "domain name system," which acts as the directory on the Internet. When a request to access a device with a domain name is given, a DNS server is queried. It returns the IP address of the device requested, allowing for proper routing.

Internet Vocabulary contd/

- ✓ Packet-switching: When a packet is sent from one device out over the Internet, it does not follow a straight path to its destination. Instead, it is passed from one router to another across the Internet until it is reaches its destination. In fact, sometimes two packets from the same message will take different routes! Sometimes, packets will arrive at their destination out of order. When this happens, the receiving device restores them to their proper order. For more details on packet-switching, see this interactive web page.
- ✓ Protocol: In computer networking, a protocol is the set of rules that allow two (or more) devices to exchange information back and forth across the network.